Results for algebra2005

theta_m=0.8, theta_f=0.8, rec=50 | N=5,10,...,50 | Ep thresholds 0.4, 0.5, 0.6, 0.7

Each table ranks models within the same graph and student split, using unrounded values.
Bold = best; underline = second best. Ties at the best value are all bold.

Experimental groups

Group	Models	exfr train edges	Test students
1. Complete forgetting graph (theta_m=0.8, theta_f=0.8, rec=50)	8/8	466,120	144
2. Remove zero-forgetting edges	6/6	258,120	144
3. Weighted nonzero sampling, up to 400/student	6/6	146,875	144
4. Weighted nonzero sampling, up to 200/student	6/6	75,932	144

Model columns

2CKG4ER, No-mastery, No-forgetting, No-student-feature, No-exercise-feature, ID-based only

TransE and TransE-adv appear only in groups with eight completed models.

Source locations

_sensitivity_runs/ + runs_Data_Fin_New_comparison/

_forgetting_edge_runs_rec50/drop_exfr0p00/

_forgetting_edge_runs_rec50/weighted_nonzero_400/

_forgetting_edge_runs_rec50/weighted_nonzero_200/

Group 1: Complete forgetting graph (theta_m=0.8, theta_f=0.8, rec=50)

Train edges: mastery 48,160; forgetting 466,120; recommendation 21,500 | 8/8 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	**0.774**	0.468	0.699	0.738	0.767	0.678	0.757	0.770
10	0.746	0.416	0.699	0.715	0.746	0.661	0.742	**0.748**
15	**0.736**	0.421	0.699	0.697	0.732	0.649	0.720	0.723
20	**0.723**	0.417	0.699	0.678	0.715	0.635	0.699	0.699
25	0.692	0.426	**0.702**	0.668	0.694	0.625	0.674	0.665
30	0.670	0.423	**0.708**	0.656	0.667	0.614	0.656	0.644
35	0.656	0.430	**0.711**	0.644	0.651	0.607	0.642	0.618
40	0.649	0.422	**0.715**	0.634	0.645	0.596	0.637	0.599
45	0.643	0.413	**0.716**	0.621	0.641	0.587	0.634	0.584
50	0.641	0.402	**0.715**	0.610	0.637	0.579	0.630	0.568

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.990	0.975	0.991	0.990	**0.991**	0.981	0.986	0.991
10	0.988	0.974	**0.991**	0.988	0.990	0.982	0.986	0.990
15	0.987	0.975	**0.991**	0.989	0.986	0.984	0.988	0.989
20	0.986	0.975	**0.991**	0.988	0.985	0.985	0.985	0.986
25	0.985	0.976	**0.991**	0.988	0.985	0.984	0.985	0.985
30	0.985	0.975	**0.991**	0.988	0.984	0.984	0.984	0.986
35	0.984	0.976	**0.991**	0.988	0.983	0.984	0.984	0.986
40	0.984	0.976	**0.991**	0.988	0.983	0.983	0.984	0.986
45	0.984	0.975	**0.991**	0.988	0.983	0.984	0.984	0.985
50	0.985	0.975	**0.990**	0.988	0.984	0.984	0.984	0.985

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 1: Complete forgetting graph (theta_m=0.8, theta_f=0.8, rec=50)

Train edges: mastery 48,160; forgetting 466,120; recommendation 21,500 | 8/8 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.132	0.023	**0.169**	-0.077	-0.135	0.040	-0.059	-0.147
10	0.107	0.068	**0.233**	-0.012	0.051	0.064	0.055	0.039
15	0.033	0.124	**0.246**	0.043	0.031	0.080	0.053	0.031
20	-0.032	0.116	**0.263**	0.082	-0.061	0.089	0.102	0.040
25	-0.182	0.126	**0.161**	0.088	-0.112	0.068	0.030	0.041
30	0.018	0.059	-0.083	0.103	-0.002	0.032	**0.134**	0.096
35	0.193	0.004	0.010	0.126	**0.200**	0.088	0.179	0.094
40	0.224	0.009	0.023	0.143	**0.245**	0.098	0.184	0.135
45	0.252	-0.029	0.100	0.138	**0.260**	0.095	0.170	0.166
50	**0.224**	0.025	0.195	0.145	0.202	0.071	0.180	0.205

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.186	0.006	**0.166**	-0.095	-0.162	0.025	-0.078	-0.158
10	0.074	0.046	**0.231**	-0.060	0.037	0.050	-0.076	-0.006
15	0.024	0.111	**0.245**	-0.002	0.027	0.069	-0.076	-0.003
20	-0.033	0.108	**0.263**	0.043	-0.062	0.075	0.083	0.023
25	-0.226	0.122	**0.160**	0.063	-0.124	0.044	0.017	0.033
30	-0.102	0.050	-0.101	0.085	-0.030	-0.001	**0.114**	0.085
35	0.167	-0.023	-0.004	0.105	**0.190**	0.053	0.170	0.078
40	0.211	-0.016	0.021	0.122	**0.241**	0.056	0.175	0.115
45	0.244	-0.055	0.099	0.107	**0.255**	0.057	0.163	0.146
50	**0.218**	0.007	0.195	0.112	0.195	0.032	0.166	0.186

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 1: Complete forgetting graph (theta_m=0.8, theta_f=0.8, rec=50)

Train edges: mastery 48,160; forgetting 466,120; recommendation 21,500 | 8/8 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.318	-0.053	**0.112**	-0.201	-0.281	-0.030	-0.201	-0.267
10	-0.037	-0.050	**0.174**	-0.192	-0.100	-0.038	-0.191	-0.131
15	-0.059	0.083	**0.204**	-0.155	-0.103	-0.032	-0.184	-0.091
20	-0.150	0.075	**0.228**	-0.115	-0.226	-0.031	-0.011	-0.071
25	-0.409	0.083	**0.124**	-0.102	-0.371	-0.073	-0.125	-0.103
30	-0.302	**-0.027**	-0.253	-0.079	-0.317	-0.128	-0.054	-0.086
35	0.052	-0.093	-0.353	-0.059	**0.066**	-0.095	0.064	-0.050
40	0.129	-0.065	-0.393	-0.032	**0.177**	-0.093	0.093	0.025
45	0.191	-0.121	-0.338	-0.027	**0.210**	-0.086	0.085	0.079
50	**0.172**	-0.053	-0.291	-0.021	0.136	-0.111	0.076	0.125

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.388	-0.160	**-0.009**	-0.268	-0.378	-0.098	-0.277	-0.327
10	-0.103	-0.148	**0.011**	-0.295	-0.273	-0.140	-0.271	-0.229
15	-0.273	-0.017	**0.006**	-0.266	-0.259	-0.175	-0.267	-0.237
20	-0.332	-0.029	**0.010**	-0.257	-0.424	-0.203	-0.141	-0.223
25	-0.451	**-0.024**	-0.178	-0.249	-0.446	-0.257	-0.269	-0.258
30	-0.344	**-0.184**	-0.455	-0.248	-0.392	-0.302	-0.195	-0.226
35	**-0.016**	-0.262	-0.414	-0.226	-0.104	-0.261	-0.144	-0.223
40	**0.081**	-0.265	-0.409	-0.219	0.072	-0.262	-0.102	-0.163
45	**0.137**	-0.284	-0.345	-0.217	0.136	-0.271	-0.104	-0.071
50	**0.121**	-0.246	-0.295	-0.210	0.038	-0.292	-0.111	-0.022

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 2: Remove zero-forgetting edges

Train edges: mastery 48,160; forgetting 258,120; recommendation 21,500 | 6/6 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.789**	0.451	0.699	0.725	0.768	0.699
10	**0.746**	0.454	0.699	0.713	0.743	0.683
15	**0.728**	0.463	0.699	0.697	0.726	0.667
20	**0.725**	0.428	0.699	0.688	0.713	0.650
25	0.695	0.413	**0.700**	0.672	0.694	0.639
30	0.680	0.409	**0.708**	0.659	0.673	0.627
35	0.674	0.397	**0.712**	0.646	0.662	0.618
40	0.662	0.392	**0.715**	0.638	0.655	0.609
45	0.654	0.384	**0.716**	0.627	0.651	0.600
50	0.652	0.374	**0.715**	0.617	0.650	0.592

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.993**	0.975	0.991	0.992	0.989	0.987
10	0.987	0.975	0.991	**0.993**	0.985	0.986
15	0.985	0.975	0.991	**0.992**	0.984	0.986
20	0.984	0.973	0.991	**0.992**	0.984	0.987
25	0.985	0.973	**0.991**	0.991	0.984	0.988
30	0.985	0.974	**0.991**	0.990	0.984	0.988
35	0.986	0.973	0.991	**0.991**	0.984	0.988
40	0.985	0.973	**0.991**	0.990	0.985	0.988
45	0.985	0.973	**0.991**	0.990	0.985	0.988
50	0.985	0.972	**0.990**	0.989	0.985	0.988

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 2: Remove zero-forgetting edges

Train edges: mastery 48,160; forgetting 258,120; recommendation 21,500 | 6/6 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.192	-0.049	**0.176**	-0.064	-0.072	0.016
10	0.021	0.113	**0.233**	-0.009	0.028	0.039
15	0.109	0.121	**0.243**	0.046	0.060	0.088
20	0.069	0.072	**0.259**	0.084	0.010	0.088
25	-0.108	0.104	**0.203**	0.116	-0.058	0.094
30	-0.032	0.079	0.015	**0.121**	0.114	0.120
35	0.105	0.099	-0.016	**0.132**	0.055	0.117
40	**0.187**	0.084	-0.027	0.143	0.162	0.125
45	**0.232**	0.090	0.091	0.147	0.207	0.126
50	**0.232**	0.082	0.124	0.135	0.222	0.115

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.195	-0.143	**0.172**	-0.080	-0.084	-0.004
10	0.013	0.067	**0.232**	-0.066	0.002	0.010
15	0.106	0.101	**0.242**	-0.012	0.057	0.058
20	0.068	0.054	**0.258**	0.047	0.009	0.065
25	-0.111	0.075	**0.202**	0.092	-0.079	0.071
30	-0.069	0.059	0.015	0.102	**0.110**	0.098
35	0.087	0.076	-0.028	**0.114**	0.053	0.097
40	**0.175**	0.064	-0.029	0.131	0.160	0.106
45	**0.224**	0.074	0.091	0.128	0.206	0.104
50	**0.226**	0.074	0.124	0.114	0.221	0.089

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 2: Remove zero-forgetting edges

Train edges: mastery 48,160; forgetting 258,120; recommendation 21,500 | 6/6 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.310	-0.244	**0.113**	-0.181	-0.224	-0.082
10	-0.207	-0.001	**0.177**	-0.207	-0.165	-0.102
15	0.001	0.014	**0.193**	-0.179	-0.035	-0.077
20	0.025	-0.032	**0.209**	-0.128	-0.100	-0.079
25	-0.260	0.024	**0.156**	-0.071	-0.265	-0.063
30	-0.320	**0.007**	-0.127	-0.054	-0.109	-0.019
35	-0.215	**0.011**	-0.350	-0.031	-0.118	-0.027
40	**0.042**	0.014	-0.412	-0.026	0.026	-0.040
45	**0.136**	0.032	-0.360	-0.011	0.111	-0.042
50	**0.157**	0.043	-0.342	-0.015	0.149	-0.058

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.377	-0.284	**-0.003**	-0.264	-0.317	-0.189
10	-0.316	-0.166	**0.006**	-0.320	-0.276	-0.239
15	-0.218	-0.136	**-0.004**	-0.291	-0.268	-0.242
20	-0.298	-0.162	**-0.002**	-0.261	-0.333	-0.268
25	-0.450	-0.078	**-0.074**	-0.235	-0.439	-0.264
30	-0.422	**-0.116**	-0.402	-0.233	-0.268	-0.236
35	-0.327	**-0.118**	-0.429	-0.224	-0.311	-0.245
40	-0.189	**-0.141**	-0.428	-0.221	-0.251	-0.238
45	**-0.009**	-0.145	-0.367	-0.217	-0.184	-0.236
50	**0.053**	-0.129	-0.345	-0.222	-0.164	-0.250

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 3: Weighted nonzero sampling, up to 400/student

Train edges: mastery 48,160; forgetting 146,875; recommendation 21,500 | 6/6 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.791**	0.486	0.699	0.712	0.762	0.696
10	**0.754**	0.492	0.699	0.705	0.749	0.680
15	**0.736**	0.450	0.699	0.690	0.732	0.670
20	**0.727**	0.426	0.699	0.680	0.721	0.662
25	**0.719**	0.409	0.699	0.672	0.700	0.658
30	0.698	0.405	**0.706**	0.666	0.682	0.648
35	0.690	0.408	**0.712**	0.659	0.668	0.640
40	0.685	0.409	**0.715**	0.651	0.656	0.631
45	0.679	0.396	**0.716**	0.642	0.653	0.622
50	0.666	0.395	**0.715**	0.634	0.649	0.612

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.994**	0.976	0.991	0.990	0.991	0.987
10	0.989	0.976	**0.991**	0.990	0.988	0.986
15	0.986	0.974	**0.991**	0.990	0.985	0.986
20	0.986	0.973	**0.991**	0.990	0.985	0.986
25	0.986	0.973	**0.991**	0.989	0.985	0.987
30	0.986	0.973	**0.991**	0.988	0.985	0.986
35	0.987	0.973	**0.991**	0.988	0.985	0.987
40	0.987	0.974	**0.991**	0.988	0.985	0.987
45	0.987	0.973	**0.991**	0.988	0.985	0.987
50	0.987	0.972	**0.990**	0.987	0.985	0.987

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 3: Weighted nonzero sampling, up to 400/student

Train edges: mastery 48,160; forgetting 146,875; recommendation 21,500 | 6/6 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.144	0.030	**0.167**	-0.035	-0.065	0.008
10	0.082	0.055	**0.243**	0.042	0.006	0.059
15	0.098	0.068	**0.245**	0.093	0.081	0.068
20	0.116	0.148	**0.256**	0.122	0.082	0.099
25	0.196	0.091	**0.246**	0.133	-0.030	0.093
30	-0.052	**0.139**	-0.060	0.130	-0.104	0.105
35	-0.025	0.083	-0.053	**0.143**	0.128	0.120
40	0.054	0.107	-0.049	0.129	**0.209**	0.125
45	0.171	0.118	0.094	0.111	**0.177**	0.121
50	0.106	0.061	0.108	0.111	**0.142**	0.118

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.152	-0.034	**0.163**	-0.056	-0.119	-0.006
10	0.049	-0.005	**0.242**	0.005	-0.023	0.036
15	0.092	0.034	**0.244**	0.076	0.074	0.052
20	0.114	0.116	**0.255**	0.110	0.081	0.086
25	0.196	0.049	**0.246**	0.119	-0.055	0.079
30	-0.087	**0.117**	-0.060	0.116	-0.115	0.089
35	-0.045	0.060	-0.054	**0.129**	0.125	0.103
40	0.039	0.091	-0.050	0.115	**0.207**	0.109
45	0.164	0.109	0.094	0.097	**0.175**	0.100
50	0.102	0.057	0.108	0.089	**0.139**	0.100

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 3: Weighted nonzero sampling, up to 400/student

Train edges: mastery 48,160; forgetting 146,875; recommendation 21,500 | 6/6 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.292	-0.154	**0.106**	-0.147	-0.249	-0.092
10	-0.151	-0.179	**0.187**	-0.123	-0.194	-0.085
15	0.021	-0.143	**0.199**	-0.070	-0.047	-0.086
20	0.025	-0.002	**0.223**	-0.024	-0.002	-0.066
25	0.133	-0.067	**0.217**	-0.003	-0.259	-0.068
30	-0.209	**0.067**	-0.199	-0.002	-0.322	-0.071
35	-0.276	**0.019**	-0.334	0.002	-0.112	-0.037
40	-0.281	0.059	-0.417	-0.009	**0.114**	-0.037
45	-0.089	0.089	-0.360	-0.035	**0.110**	-0.035
50	-0.058	0.028	-0.349	-0.044	**0.084**	-0.026

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.348	-0.197	**-0.009**	-0.215	-0.348	-0.199
10	-0.284	-0.260	**0.022**	-0.225	-0.327	-0.224
15	-0.169	-0.232	**0.006**	-0.198	-0.264	-0.250
20	-0.252	-0.099	**0.005**	-0.184	-0.302	-0.244
25	-0.208	-0.161	**-0.015**	-0.204	-0.409	-0.262
30	-0.379	**-0.020**	-0.412	-0.219	-0.459	-0.267
35	-0.387	**-0.097**	-0.459	-0.212	-0.256	-0.251
40	-0.350	-0.067	-0.438	-0.220	**-0.040**	-0.240
45	-0.241	**-0.011**	-0.370	-0.239	-0.108	-0.251
50	-0.211	**-0.140**	-0.354	-0.242	-0.181	-0.257

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 4: Weighted nonzero sampling, up to 200/student

Train edges: mastery 48,160; forgetting 75,932; recommendation 21,500 | 6/6 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.772**	0.298	0.699	0.699	0.763	0.712
10	0.744	0.344	0.699	0.686	**0.746**	0.698
15	**0.735**	0.374	0.699	0.676	0.731	0.685
20	**0.723**	0.401	0.699	0.664	0.710	0.672
25	0.690	0.405	**0.699**	0.659	0.684	0.661
30	0.662	0.423	**0.706**	0.650	0.664	0.651
35	0.652	0.425	**0.712**	0.641	0.651	0.644
40	0.637	0.425	**0.713**	0.632	0.641	0.636
45	0.639	0.415	**0.715**	0.621	0.635	0.627
50	0.632	0.407	**0.715**	0.613	0.628	0.619

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.988	0.969	**0.991**	0.986	0.987	0.987
10	0.984	0.969	**0.991**	0.987	0.988	0.988
15	0.984	0.970	**0.991**	0.985	0.984	0.989
20	0.984	0.971	**0.991**	0.986	0.983	0.989
25	0.984	0.972	**0.991**	0.986	0.984	0.988
30	0.983	0.972	**0.991**	0.986	0.983	0.989
35	0.983	0.972	**0.991**	0.986	0.983	0.988
40	0.983	0.972	**0.991**	0.986	0.983	0.988
45	0.983	0.972	**0.991**	0.987	0.983	0.988
50	0.984	0.972	**0.990**	0.987	0.984	0.989

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 4: Weighted nonzero sampling, up to 200/student

Train edges: mastery 48,160; forgetting 75,932; recommendation 21,500 | 6/6 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.026	0.002	**0.163**	-0.041	-0.057	0.009
10	0.036	0.057	**0.230**	0.003	-0.007	0.058
15	0.052	0.059	**0.245**	0.072	0.086	0.075
20	-0.002	0.086	**0.267**	0.084	-0.053	0.100
25	-0.185	0.106	**0.246**	0.101	-0.084	0.109
30	-0.072	0.082	0.051	0.125	0.112	**0.126**
35	**0.197**	0.087	-0.050	0.116	0.150	0.118
40	**0.226**	0.082	0.075	0.133	0.185	0.124
45	**0.273**	0.094	-0.018	0.144	0.166	0.143
50	**0.170**	0.095	0.121	0.142	0.126	0.152

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.044	-0.093	**0.158**	-0.066	-0.076	-0.010
10	0.033	0.021	**0.228**	-0.030	-0.043	0.036
15	0.051	0.040	**0.244**	0.039	0.071	0.054
20	-0.002	0.059	**0.266**	0.062	-0.058	0.087
25	-0.219	0.083	**0.246**	0.084	-0.132	0.093
30	-0.155	0.072	0.050	0.108	0.064	**0.115**
35	**0.169**	0.081	-0.050	0.101	0.135	0.103
40	**0.211**	0.073	0.075	0.117	0.178	0.103
45	**0.263**	0.086	-0.018	0.127	0.161	0.125
50	**0.163**	0.091	0.121	0.120	0.114	0.127

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 4: Weighted nonzero sampling, up to 200/student

Train edges: mastery 48,160; forgetting 75,932; recommendation 21,500 | 6/6 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.220	-0.193	**0.096**	-0.161	-0.203	-0.092
10	-0.101	-0.033	**0.169**	-0.156	-0.292	-0.076
15	-0.059	-0.017	**0.196**	-0.103	-0.104	-0.076
20	-0.080	0.010	**0.229**	-0.091	-0.171	-0.050
25	-0.361	0.020	**0.211**	-0.079	-0.339	-0.045
30	-0.350	**0.025**	-0.120	-0.057	-0.170	-0.019
35	0.029	**0.047**	-0.279	-0.066	0.031	-0.031
40	**0.150**	0.039	-0.333	-0.044	0.122	-0.034
45	**0.226**	0.060	-0.416	-0.021	0.114	-0.012
50	**0.129**	0.066	-0.357	-0.034	0.065	-0.004

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.299	-0.233	**-0.017**	-0.232	-0.314	-0.198
10	-0.293	-0.122	**0.002**	-0.277	-0.381	-0.226
15	-0.310	-0.119	**-0.005**	-0.246	-0.237	-0.246
20	-0.299	-0.088	**0.002**	-0.239	-0.381	-0.241
25	-0.480	-0.091	**-0.028**	-0.242	-0.432	-0.248
30	-0.412	**-0.157**	-0.344	-0.216	-0.285	-0.228
35	**-0.108**	-0.142	-0.455	-0.240	-0.184	-0.252
40	**0.072**	-0.162	-0.384	-0.217	-0.051	-0.249
45	**0.160**	-0.110	-0.430	-0.202	-0.118	-0.223
50	**-0.043**	-0.114	-0.364	-0.210	-0.233	-0.219

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.